

What moves *the price.*

Premium, decoded
Intrinsic + extrinsic
Time · Volatility
Right, and still losing

One page

Read after
No. 01 & No. 02

An option's price isn't a single mysterious number. The **premium has an anatomy** — two parts, pushed around by two forces: *time* and *volatility*. Once you can see those parts, the most confusing beginner experience finally makes sense: how you can call the direction correctly and **still lose money**.

01 /
TWO
PARTS

Every premium is two parts.

Whatever you pay for an option splits cleanly in two. One part is the value it would have if you exercised it this instant. The other is everything you're paying for the *chance* that it becomes worth more before it expires.

A \$7.00 PREMIUM · CALL, STRIKE \$100, STOCK AT \$105

Intrinsic · \$5.00

WHAT IT'S WORTH IF EXERCISED NOW

Extrinsic (time value) · \$2.00

WHAT YOU PAY FOR "MAYBE"

THE SOLID PART

● Intrinsic value

The real, exercise-it-now payoff. A call struck at \$100 with the stock at \$105 has \$5 of intrinsic — you could buy at \$100 and sell at \$105. Out-of-the-money options have **zero** intrinsic.

THE EVAPORATING PART

● Extrinsic (time) value

Everything above intrinsic — the price of possibility. It's largest when there's lots of time and lots of uncertainty, and it melts to exactly **zero** at expiration. This is the part most beginners are really buying.

02 /
INTRINSIC

Intrinsic value — worth right now.

The easy half. It's just the moneyness payoff, and it can never be negative — if exercising would lose money, you simply wouldn't, so the floor is zero.

THE FORMULA

CALL intrinsic = max(0, spot - strike)
PUT intrinsic = max(0, strike - spot)

At the moment of expiration, **all that's left is intrinsic** — every scrap of time value is gone. That's why an option's expiry value is fully determined by where the stock lands.

CALL STRUCK AT \$100

STOCK	MONEYNES	INTRINSIC
\$120	In the money	\$20
\$105	In the money	\$5
\$100	At the money	\$0
\$95	Out of the money	\$0

03 /
EXTRINSIC

Extrinsic value — *the price of "maybe."*

The interesting half, and the one that trips people up. Extrinsic value is what the market charges for the possibility that an option ends up worth more than it's worth today. Two things inflate it: **time remaining** and **expected movement (volatility)**.

MORE TIME

More days = more chances for a favourable move = more extrinsic value. A 90-day option costs more than a 7-day one, all else equal.

MORE VOLATILITY

A jumpier stock could travel further, so its options carry fatter extrinsic value. Calm stocks have cheap options.

THE CATCH

At-the-money options are **almost entirely extrinsic**. Buy one and you're mostly buying time value — which is guaranteed to decay.

The reframe.

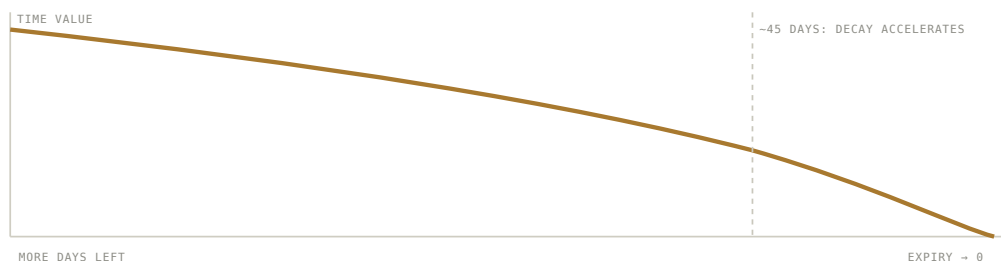
When you **buy** an option you are usually long time value — you need the stock to move enough, fast enough, to outrun the melt. When you **sell** one (covered call, cash-secured put) you are short time value — you're being paid to let it melt in your favour. Same coin, two sides — exactly as in No. 01.

04 /
TIME
DECAY

Time decay melts it — *fastest at the end.*

Extrinsic value doesn't drain at a steady drip. It clings on early and then falls off a cliff in the final weeks. This accelerating melt is called **theta**, and it's the single force working against every option buyer, every single day — weekends included.

HOW AN AT-THE-MONEY OPTION'S TIME VALUE DECAYS TOWARD EXPIRY



IF YOU BOUGHT THE OPTION

Theta is a headwind. Every day the stock does nothing, you lose a little — and you lose it **faster** as expiry nears. Give yourself time: longer-dated options decay more gently and forgive slow timing.

IF YOU SOLD THE OPTION

Theta is a tailwind. As the writer of a covered call or cash-secured put, decay is exactly what you're harvesting — the option you sold gets cheaper to buy back with each passing day.

Practical takeaway: buyers should avoid short-dated, at-the-money options as a default — you're standing directly under the fastest part of the melt. Sellers, conversely, often prefer that same 30–45 day window because the decay they collect is richest there.

05 /
VOL

Volatility sets the temperature.

If time is the slow melt, **implied volatility** is the thermostat — it sets how expensive every option is right now. IV is the market's collective guess at how much the stock will move going forward, baked straight into the premium. It is *not* a direction; a high-IV stock is simply expected to swing hard, either way.

SAME OPTION, TWO VOLATILITY REGIMES — THE EXTRINSIC VALUE INFLATES

CALL · STRIKE \$100 · STOCK \$100 · 30 DAYS · INTRINSIC = \$0, SO IT'S ALL TIME VALUE

LOW IV (20%)		\$2.30
HIGH IV (60%)		\$6.80

Nothing changed but the market's expectation of movement — yet the option costs nearly 3× as much. Buy when IV is high and you've overpaid for the "maybe"; that premium can deflate even if the stock cooperates.

IV RISES INTO EVENTS

Ahead of earnings or big news, uncertainty spikes and options get expensive — the market is bracing for a move.

IV CRUSH AFTER

Once the news is out, uncertainty vanishes and IV collapses. Premiums deflate fast — the classic trap for event-buyers.

IV RANK / PERCENTILE

A gauge of whether today's IV is high or low *for this stock*. Buyers prefer low; sellers prefer high. Most platforms show it.

Rule of thumb.

Think about IV *before* you trade, not after. As a beginner: be wary of buying options when volatility is unusually high (especially right before earnings), and remember that a covered call or cash-secured put actually **benefits** from selling that richer premium.

06 /
THE
TRAP

Why you can be right — *and still lose.*

This is the lesson the anatomy above is really for. Three forces can quietly beat you even when the stock moves your way. Usually it's a mix of them.

TRAP 1 · IV CRUSH

You bought before earnings, the stock rose — but IV collapsed and drained more value than the move added. Right direction, lower price.

TRAP 2 · TOO SLOW

The move came, but late. Time decay ate the premium while you waited for it. Being right eventually isn't the same as being right in time.

TRAP 3 · NOT ENOUGH

The stock rose, but not past your **breakeven** (strike + premium). A small favourable move can still leave a buyer underwater.

A WORKED EXAMPLE

You pay **\$3.00** for a \$100 call with 30 days left; breakeven is **\$103**. A week later the stock has ticked up from \$100 to \$102 — you were right, it rose. But it's still below breakeven, seven days of decay are gone, and IV eased off. The call is now worth **\$2.60**. Correct call, $-\$0.40$. To *profit*, you needed a move that was bigger, faster, or into cheaper volatility. This is the everyday reality options buyers must plan around.

07 /
TERMS

Words you'll meet.

Intrinsic value

Exercise-now worth: $\max(0, \text{spot} - \text{strike})$ for calls, mirror for puts. All that remains at expiry.

Extrinsic / time value

Premium above intrinsic — the price of possibility. Decays to zero by expiration.

Theta

The rate of time decay per day. Negative for buyers, positive for sellers; accelerates near expiry.

Implied volatility (IV)

The market's expected future movement, priced into the option. Higher IV = pricier options.

Historical volatility

How much the stock actually moved in the past — a reality check against implied.

IV crush

The sharp drop in IV (and premiums) once an anticipated event passes.

IV rank / percentile

Where current IV sits versus this stock's own recent range. A high/low gauge.

Breakeven

The stock price at which a trade neither makes nor loses at expiry. Long call: $\text{strike} + \text{premium}$.

Extrinsic burn

Informal: the steady loss of time value a buyer suffers just by holding.